

What the Colors Mean in the Pythagorean Radial-Gap Plot

The color represents an exact arithmetic label attached to each primitive Pythagorean triangle. It does *not* represent the triangle's size or a dynamical property of the three-body trajectory.

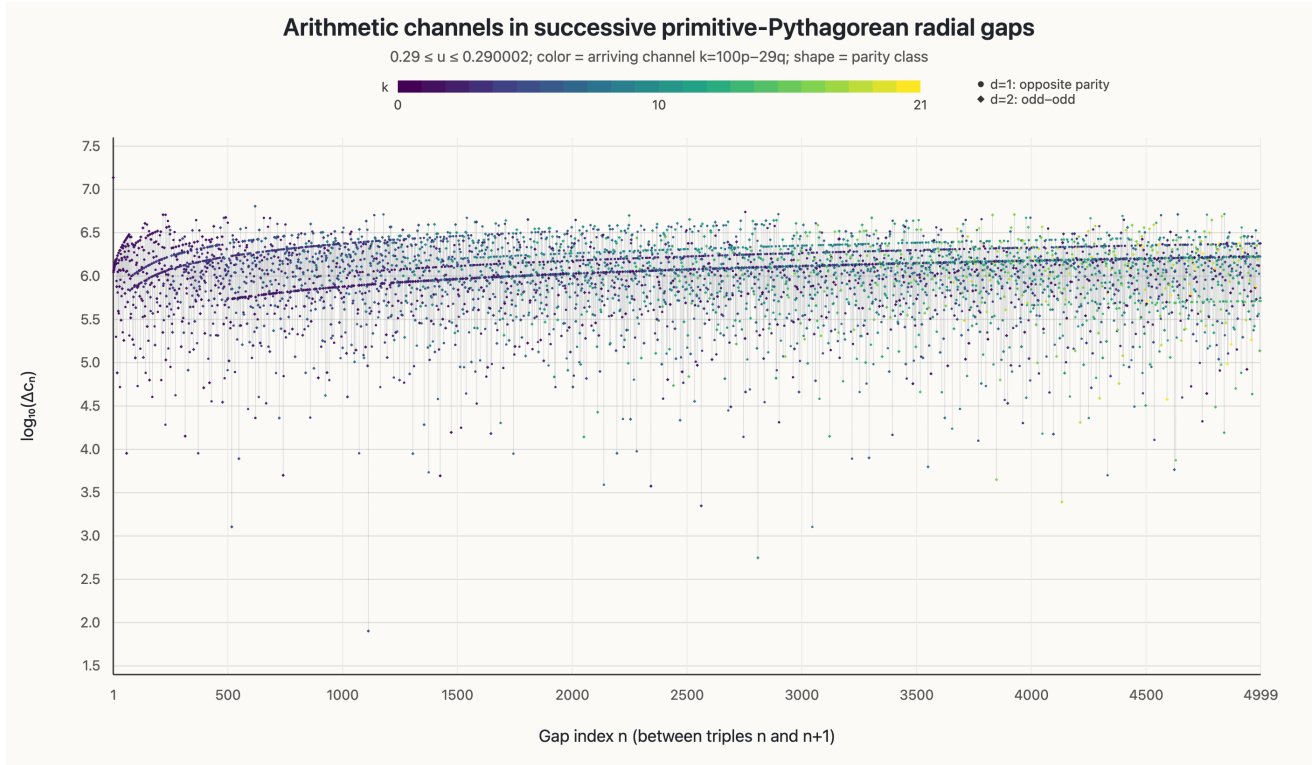


Figure 1: Successive radial gaps. Each point is colored using the arriving triangle's arithmetic channel.

1. From an actual triangle to its color

Let

$$(x, y, c), \quad x < y, \quad x^2 + y^2 = c^2,$$

be one of the primitive triples in the narrow wedge. Its Euclid parameter can be recovered directly from the three side lengths:

$$u = \frac{x}{c + y}.$$

Reduce this rational number to lowest terms, writing

$$u = \frac{p}{q}, \quad \gcd(p, q) = 1.$$

The color is determined by the integer

$$k = 100p - 29q.$$

At horizontal index n , the plotted height is

$$\log_{10}(c_{n+1} - c_n),$$

and its color is the value of k belonging to the *arriving* triangle $(x_{n+1}, y_{n+1}, c_{n+1})$.

2. Geometric meaning of the channel number

The lower edge of the parameter interval is $u = 29/100$. Therefore

$$\frac{p}{q} - \frac{29}{100} = \frac{100p - 29q}{100q} = \frac{k}{100q}.$$

Thus k labels parallel rows of rational lattice points immediately above the lower boundary of the wedge. Equivalently, it is the integer determinant

$$k = \det \begin{pmatrix} 100 & q \\ 29 & p \end{pmatrix}.$$

The color scale in this particular 5,000-triple plot is approximately

Color	Channel number
 Purple	$k \simeq 0$
 Blue	small k
 Teal	$k \simeq 10$
 Green	$k \simeq 15$
 Yellow	$k \simeq 21$

For example, the first triangle is

$$(5800, 9159, 10841).$$

It has

$$u = \frac{5800}{10841 + 9159} = \frac{29}{100},$$

and hence $k = 0$. It lies exactly on the lower endpoint of the wedge and is colored purple.

3. Why blue is gradually joined by green

The upper edge of the interval is

$$u \leq \frac{29}{100} + \frac{1}{500000}.$$

Using $u - 29/100 = k/(100q)$, this becomes

$$k \leq \frac{q}{5000}.$$

Small triangles have small Euclid denominators q , so only small channel numbers are possible. For example,

q	largest possible k
5,000	1
50,000	10
100,000	20

The hypotenuse is approximately proportional to q^2 . Moving to the right in the plot therefore moves toward larger q , progressively allowing higher channel numbers. Purple and blue dominate at first; teal and green enter as new channels become arithmetically possible. The low- k channels have not vanished—they are increasingly interlaced with the newly available channels.

4. Meaning of the marker shape

For reduced $u = p/q$, the primitive triple is

$$(x, y, c) = \frac{1}{d} (2pq, q^2 - p^2, q^2 + p^2),$$

up to ordering the two legs, where

$$d = \begin{cases} 1, & p, q \text{ have opposite parity,} \\ 2, & p, q \text{ are both odd.} \end{cases}$$

Consequently, circles in the plot denote $d = 1$, while diamonds denote $d = 2$.

5. Channel color versus physical position in the wedge

The channel number k is not by itself the fractional position across the physical wedge. That normalized position is

$$\xi = \frac{u - 0.29}{0.000002} = \frac{5000k}{q}, \quad 0 \leq \xi \leq 1.$$

Thus two triangles with the same k but different q occupy different fractional positions within the wedge. The present plot uses k because it reveals the exact arithmetic strands. A plot colored by ξ would instead display each triangle's actual position between the two wedge boundaries.